# Real-Time Video Inference on 8× NVIDIA RTX PRO 5000

MiniMax H3 with vLLM-Omni

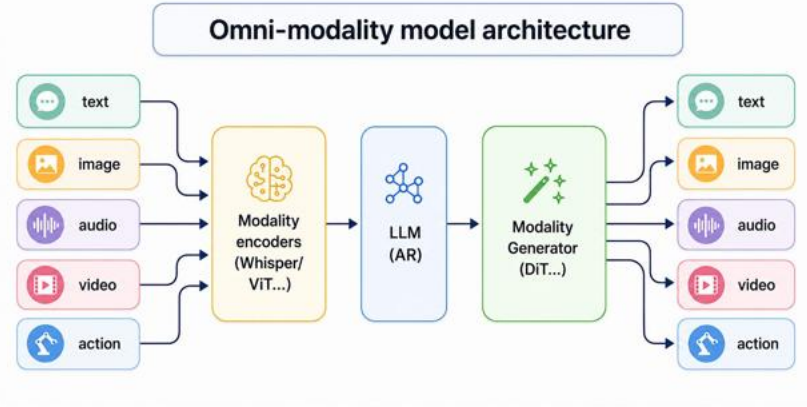
Shunyang Li | Public Technical Brief | October 2026



# vLLM-Omni Connects ComfyUI Workflows to Multi-GPU Serving

PR #9194 demonstrates MiniMax-H3 character replacement through a visual graph while stage placement, batching, and parallelism remain inside the serving runtime

## SERVING ARCHITECTURE · OFFICIAL v0.26 FIGURE



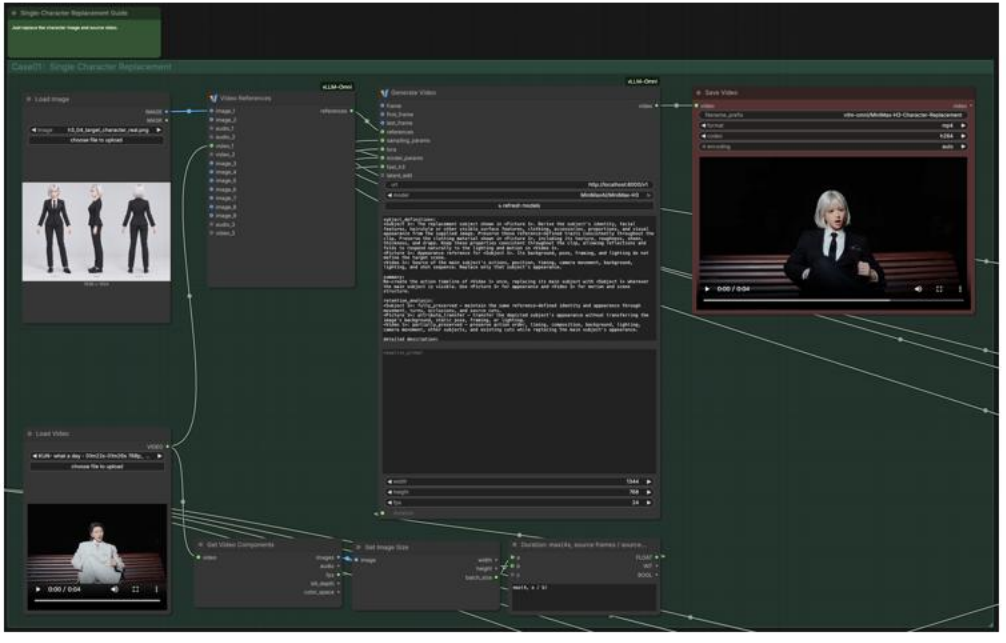

## WHAT THE SEPARATION ENABLES

**Visual authoring** — References, LoRA, sampling, and continuation stay editable in ComfyUI.

**Serving boundary** — The graph calls the vLLM-Omni Generate Video endpoint.

**Multi-GPU execution** — Placement and TP / SP / CFG / EP policy remain stage-local.

## COMFYUI WORKFLOW · PR #9194



MiniMax-H3 single-character replacement: Load Image + Load Video → Video References → vLLM-Omni Generate Video → Save Video

Public PR #9194 · open as of 24 Oct 2026



Sources: vLLM-Omni v0.26 public deck, slides 22–24 · vLLM-Omni PR #9194 (open)

# SM120 Is Compute-Rich; Fabric Bandwidth Sets Scale-Out Headroom

Ulysses SP8 moves Q/K/V and Gate/O across ranks, placing transfers and barriers on the dependency path

**PER-GPU BASELINE · EIGHT-GPU SYSTEM**

| 110 SM/GPU | 72 GB/GPU | 1,344 GB/s | 400 Gb/s |
|---|---|---|---|
| compute capability 12.0 | GDDR7 ECC | local memory / GPU | per active fabric port |

8-GPU system: 880 SM · 576 GB GPU-local memory · 8 active 400 Gb/s fabric ports · memory is not pooled

**TWO EXECUTION REGIMES**

| Path | Tensor ownership | Synchronization | Fabric pressure |
|---|---|---|---|
| **Projection / MLP / local VSA tiles** | Stays on one GPU | Few readiness fences | Modest |
| **Ulysses SP8 attention** | Q/K/V + Gate/O move across ranks | Transfers + barriers on dependency path | High |

**Communication pressure is determined by where tensors must move, not by operator FLOPs alone.**

**WHY BANDWIDTH MATTERS**

**400 Gb/s ÷ 8 = 50 GB/s**

port line rate converted to bytes

**1,344 GB/s ÷ 50 GB/s = 26.9×**

**The nominal local-memory rate is 26.9× one 400 Gb/s port's line rate; effective payload bandwidth and exposed communication therefore set the remaining scale-out headroom.**

*Raw-rate comparison; excludes protocol overhead, topology, contention, and overlap



Sources: 72 GB & 1,344 GB/s — NVIDIA RTX PRO 5000 Blackwell datasheet · 110 SM/CC 12.0 & 8×400 Gb/s — measured system inventory